

TABLE III
Some Digitisers Available in the Market

Manufacturer	Model	Features
Appicos	PD24222 digitiser	- Dual-channel, 24-bit, fully differential inputs - Input ranges from 32mV to 10V peak-to-peak and programmable DC offset from -5V to +5V - LabVIEW, Lab windows/CVI drivers and soft panels included - Suitable for audio type of measurements - Flexible input ranging and coupling allow easy adaptation to a wide range of unit under test output voltages
Keysight	M9703A wideband AXIe digitiser	- Eight channels of DC to 2GHz input frequency range, 12-bit resolution and up to 3.2GS/s sampling rate (when interleaved) - Multi-module processing synchronisation for >100 phase coherent channels - Four onboard Virtex-6 FPGAs for custom processing or OTS wideband real-time DDC - Used for multi-channel applications requiring phase coherency, wideband capability and high dynamic range, RF communications, antenna array calibration and test, electronic warfare (EW) radar emulation, passive radar receiver, multi-band SATCOM monitoring, and large scale experiments in hydrodynamics, plasma fusion (Tokamak/Stellarator), particle physics and microwave astronomy
NI (National Instruments)	NI PXIe-5170R reconfigurable digitiser/oscilloscope	- Reconfigurable FPGA for real-time decision making, hardware acceleration, high throughput processing and to serve multiple applications - Offers user reprogramming in LabVIEW FPGA or traditional HDLs - Modular PXI platform enables scaling of channels and integration with multiple devices
SP Devices	ADQ1600	14-bit single channel 1.6GSa/s digitiser in USB/PXIe/PCIe/UTCA form factors with onboard FPGA for real-time processing
Spectrum GmbH	M4i.22xx-x8 series digitiser cards	- One, two or four channels per card - Sampling rates up to 5GS/s and bandwidth up to 1GHz 4GB onboard memory and PCIe x8 Gen2 interface with transfer speed greater than 3Gbyte/s - Caters to ultrafast signals and pulses found in electronics, aerospace and defense, and science
Spectrum GmbH	DigitizerNETBOX DN2.445-04 Ethernet/LXI digitiser	- Four single-ended inputs - Sampling rates up to 500MS/s with 14-bit resolution and 250MS/s with 16-bit resolution - Programmable with SBench6 Professional, Visual C++, Borland C++, Gnu C++, Visual Basic, VB.NET, C#, J#, Delphi and Python - Suitable for applications that require high-quality signal acquisition

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Onboard memory. The speed of data transfer between a digitiser and PC through different communication buses is often less than the rate at which data is acquired by the digitiser. "In most applications that require single-shot/transient signal capture, the digitiser uses onboard memory to store the same," comments Sheikh Ahmed.

Once data is captured and stored in the memory, it is ready to be read out by the processor for local storage

and processing. The time with which the onboard memory gets filled is given by the following equation:

Buffer time = onboard memory time/(sampling rate - data transfer rate)

The equation for calculating the acquisition memory required to capture a specific duration at a particular sample rate is: as follows:

Acquisition memory = sample rate x time window

where, time window is given as time/division $\times 10$ (10 here denotes the number of divisions on the time axis.) So, to capture 1ms of signal at 1GSa/s sample rate, we require one mega sample memory. Therefore for applications where high-speed analogue-to-digital conversions are required, the memory should also be high to avoid loss of data.

What is new

Reconfigurable digitisers use field programmable gate arrays (FPGAs) for achieving real-time decision making and high-throughput processing in measurement systems. Users can reprogram the instrument and develop custom firmware to cater to different applications without needing extensive knowledge of system design, data handling and FPGA programming. "The computational element also avoids the issue of high-speed data transfer through an external bus," says Misra. ●